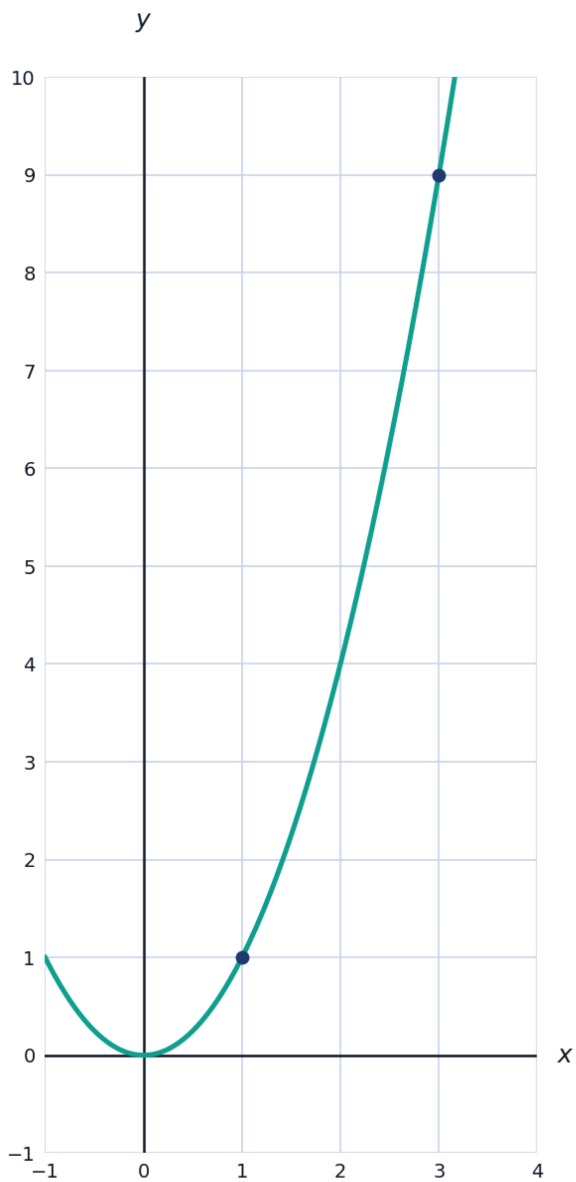
The Derivative from First Principles

The definition of the derivative

1 Use the limit definition $f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$ to find $f'(x)$ for $f(x) = x^2 + 3x$.

2 Use the limit definition to find $f'(x)$ for $f(x) = \sqrt{x}$ (for $x > 0$).

3 The graph shows $f(x) = x^2$ with the secant line through $(1, 1)$ and $(3, 9)$.



a Find the slope of the secant line.

b Find $f'(2)$ using the derivative, and compare it to the secant slope.

Differentiability

4 Explain why $f(x) = |x|$ is not differentiable at $x = 0$, using one-sided derivatives.

5 State the relationship between differentiability and continuity: if a function is differentiable at a point, must it be continuous there? Is the converse true?

More derivatives from the definition

6 Use the limit definition to find $f'(x)$ for $f(x) = \frac{1}{x}$ (for $x \neq 0$).

7 Use the limit definition to find $f'(x)$ for $f(x) = x^3$.

Interpreting the derivative

8 A ball's height is $h(t) = -5t^2 + 20t + 2$ (metres, seconds). Use the derivative definition idea to explain what $h'(2)$ represents physically, and find its value using $h'(t) = -10t + 20$.